

Chapter 32. Mechanical Ventilation in Neuromuscular Disease

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Mechanical Ventilation in Neuromuscular Disease: Introduction

The earliest application of assisted ventilation in patients with neuromuscular disease was with the iron lung during the poliomyelitis epidemic in the 1940s and 1950s, which enabled patients with severe respiratory insufficiency to recover, become ventilator-free, and go on to lead productive lives.¹ In most cases, the device was used for a few weeks to up to 2 years, with eventual recovery of ventilatory function. Many polio patients with disability continue to reside in iron lungs in the community after several decades.² Others have converted to tracheostomy-assisted positive-pressure ventilation, achieving mobility and the ability to clear airway secretions. With appropriate weaning techniques, some have switched to noninvasive positive-pressure ventilation (NIPPV).³ NIPPV is now preferred to support most patients with chronic neuromuscular disorders and is used increasingly to support patients with acute ventilatory insufficiency, such as with Guillain-Barré syndrome and myasthenia gravis. Table 32-1 lists neuromuscular conditions associated with respiratory impairment and failure.

Table 32-1: Neuromuscular Diseases Associated with Respiratory Impairment or Failure

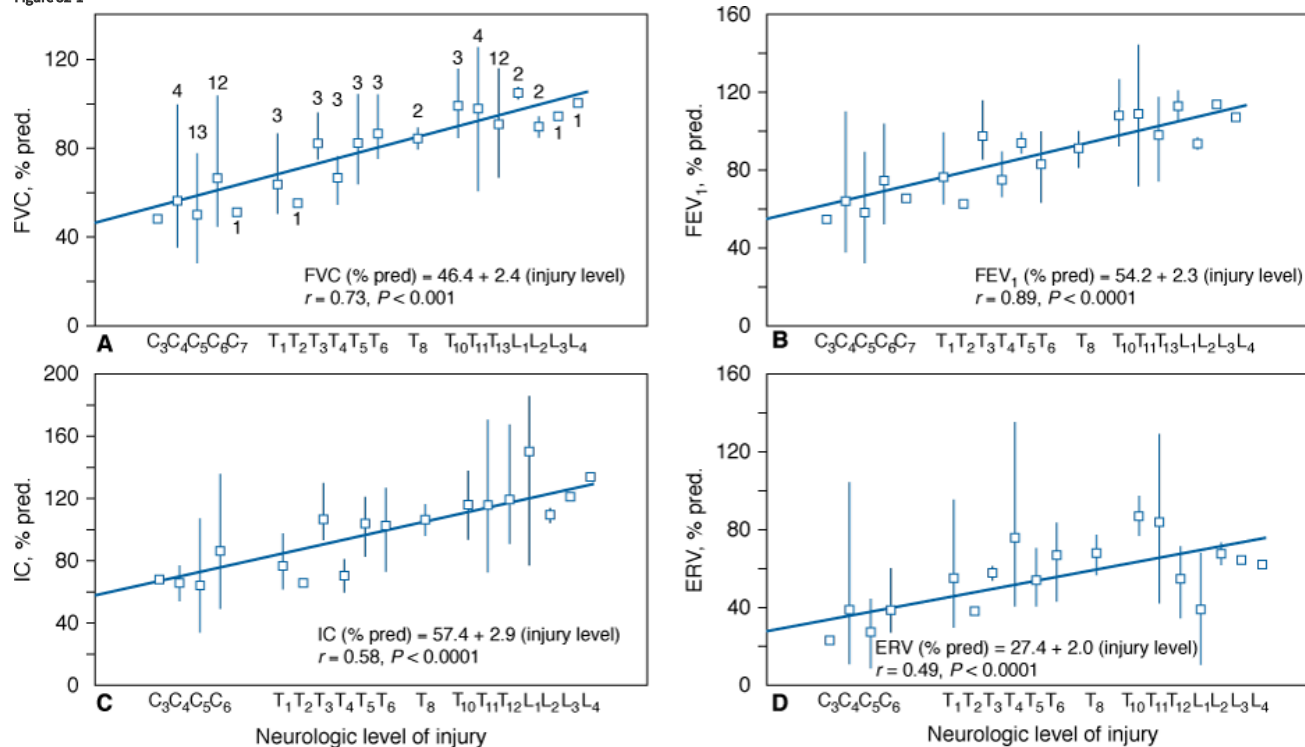
1. Muscle diseases
a. Dystrophies (Duchenne, fascioscapulohumeral, limb girdle)
b. Myotonias
c. Autoimmune/inflammatory myopathies (dermatomyositis, polymyositis)
d. Metabolic myopathies (hypophosphatemia, glycogen-storage disorders, disturbed lipid metabolism)
e. Endocrine myopathies (hypothyroidism, hyperthyroidism)
f. Periodic paralysis
g. Toxic myopathies (alcohol)
2. Peripheral neuropathies
a. Autoimmune/inflammatory (Guillain-Barré, Chronic demyelinating inflammatory polyneuropathy)
b. Toxic (heavy metal, organophosphates, nitrofurantoin, hexacarbons)
c. Acute intermittent porphyria
d. Tick paralysis
e. Shellfish poisoning
f. Hereditary (Charcot-Marie-Tooth)
g. Endocrine (hypothyroidism, hyperthyroidism, diabetic)
3. Neuromuscular junction (myasthenia gravis, botulism)
4. Motor neuron disease
a. Poliomyelitis
b. Amyotrophic lateral sclerosis
c. Spinal muscle atrophies
5. Intensive care unit-related weakness
a. Critical illness polyneuropathy (sepsis, multisystem organ failure)
b. Neuromuscular blocking agent-related neuropathy
c. Status asthmaticus-associated neuromyopathy (high-dose steroids, aminoglycosides, paralytic agents)
6. Spinal cord injury

Overview of Pertinent Pathophysiology

The primary muscle of inspiration is the diaphragm, which is innervated by the third through fifth cervical nerves. In addition, the external intercostal muscles provide stability to the rib cage during inspiration. With ascending paralysis, such as with Guillain-Barré syndrome or traumatic quadriplegia above the T10

level (Fig. 32-1),⁴ there are reductions in vital capacity and other volume subdivisions, as well as rib cage distortion with inspiratory effort. These changes result in regional ventilation-perfusion mismatching, particularly in the dependent portions of the lungs. An early increase in the alveolar-arterial oxygen (O_2) difference can be found in some patients with neuromuscular impairment long before hypercapnia develops.^{5,6} Gas exchange worsens during sleep secondary to alveolar hypoventilation, inhibited intercostal and accessory muscle activity, and dead space ventilation induced by rapid shallow breathing.⁶⁻⁸ Long-term episodic hypoxia in dystrophic mice leads to dysfunction of myofibrillar contractility as there is no change in diaphragmatic collagen content and dry-to-wet ratio.⁹ Expiratory muscles in muscular dystrophies are more impaired than inspiratory muscles.¹⁰ Despite this finding, cough effort can still be preserved if inspiratory volume and respiratory elastic recoil are maintained.¹¹

Figure 32-1



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Spirometric variables in seated spinal cord-injured subjects distributed according to lesion level. **A.** Forced vital capacity (FVC; $n = 74$). **B.** Forced expiratory volume in 1 second (FEV₁; $n = 74$). **C.** Inspiratory capacity (IC; $n = 73$). **D.** Expiratory reserve volume (ERV; $n = 73$). Values are means and ranges of percent predicted (% pred) values for healthy persons. Numbers above squares in A are numbers of subjects at each injury level and are the same for B to D. (Used, with permission, from Baydur et al.⁴)

Patients with Duchenne muscular dystrophy with a vital capacity above 1.8 L remain normocapnic.¹² Below this value, nocturnal hypercapnia is likely to ensue. Many factors contribute to hypercapnia including respiratory muscle weakness,¹³ central hypopneas during rapid-eye movement sleep,¹⁴ upper airways obstruction,¹⁵ sleep fragmentation,¹⁶ obesity,¹⁴ respiratory muscle asynchrony,¹⁷ and decrease in respiratory compliance.^{18,19} Thoracic scoliosis, often associated with disorders such as poliomyelitis and muscular dystrophy (Fig. 32-2), further reduces respiratory compliance, resulting in an increase in inspiratory drive, altered force-length and force-velocity relationships,^{20,21} and, ultimately, diaphragmatic fatigue.

Figure 32-2



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Patient with Duchenne muscular dystrophy and severe scoliosis.

Body position influences regional ventilation in individuals with neuromuscular disease. Furthermore, use of positive-pressure ventilation and an abdominal belt results in greater global and regional ventilation in the supine position than in the right and left lateral positions.²² Intermittent positive-pressure breathing sessions (30 consecutive deep breaths at pressures up to 30 cm H₂O) improve regional ventilation by approximately 30%, more so in the anterior regions.^{22,23}

Measurement of maximal static mouth pressures (P_{max}) provides a guide to the degree of respiratory muscle weakness and is an adjunct in assessing the need for assisted ventilation. In stable neuromuscular disease, maximal expiratory mouth pressures are lower in myopathy than in polyneuropathy and in proximal than in distal muscle weakness.^{10,24}

Weakened abdominal muscles reduce the ability to cough, as reflected by decreased peak expiratory flow rates. Patients with peak expiratory flows of less than 160 L/min need cough assistance to prevent accumulation of airway secretions.^{25,26} Cough impairment is worsened by inspiratory muscle weakness and glottic dysfunction. Bulbar muscle involvement impairs swallowing and speech and increases risk of aspiration and upper airway collapse.

Dysregulation of breathing in muscular dystrophy and myotonic dystrophy has been attributed to abnormal feedback from respiratory muscle receptors and abnormal central ventilatory control, respectively.^{27,28} Sleep-disordered breathing can be related to respiratory muscle weakness, chest wall deformities, obesity, craniofacial abnormalities, bulbar dysfunction, and abnormalities in control of ventilation.^{14,29} Loss of rib cage and bulbar muscle tone, particularly during rapid-eye movement sleep, results in hypoventilation, decreased inspiratory flow, fragmented sleep, and daytime symptoms, well described in myotonic dystrophy, diaphragmatic paralysis, Duchenne muscular dystrophy, and amyotrophic lateral sclerosis.^{30–34} Despite muscle weakness, central drive in patients with neuromuscular and chest wall disorders is often increased.³⁵

Clinical and Physiologic Indications to Initiate Assisted Ventilation

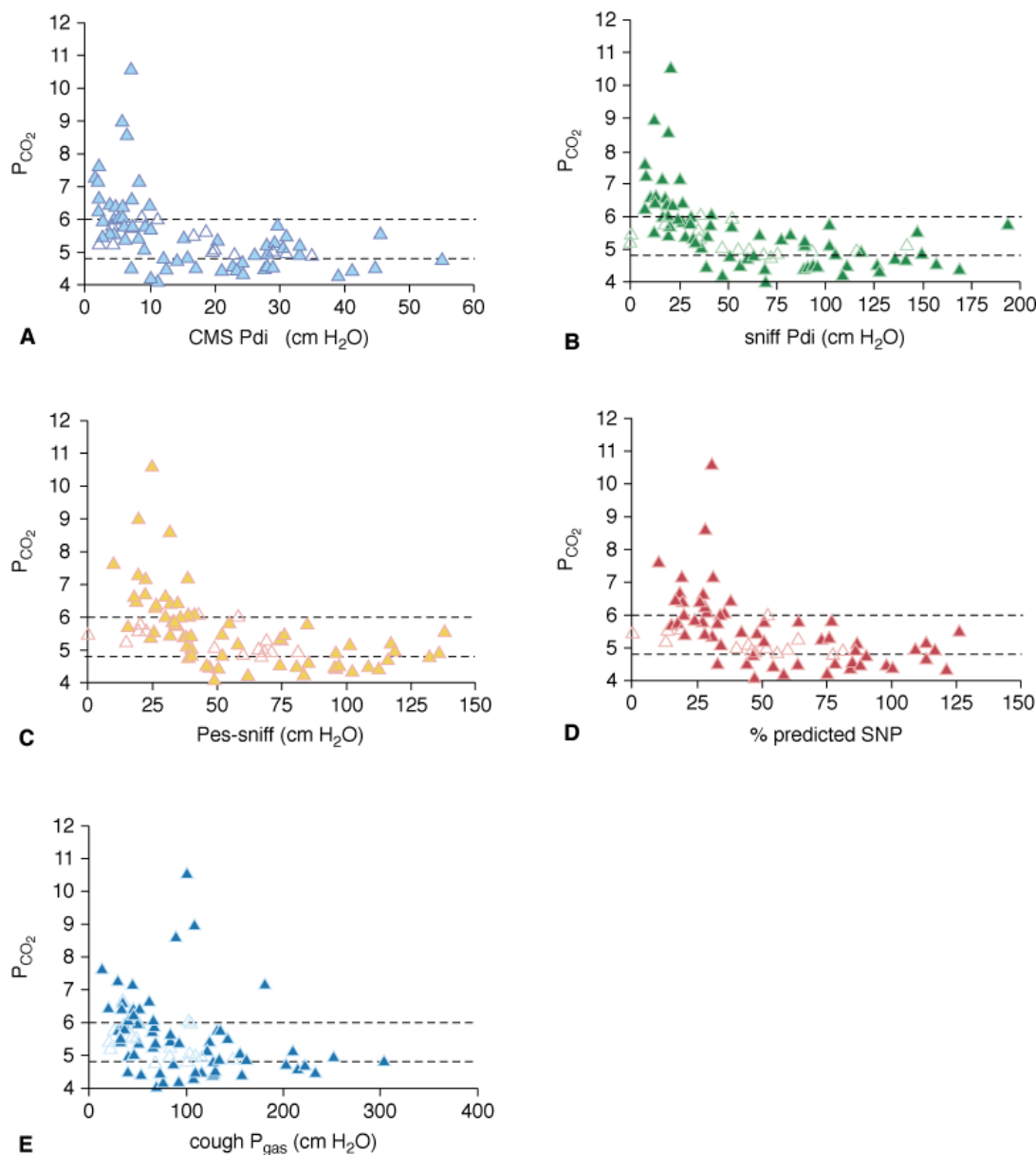
Altered mental status, cardiac or respiratory arrest, shock, arrhythmias, gas-exchange abnormalities, and bulbar dysfunction are absolute indications for endotracheal intubation. Such acute changes are frequently seen in patients with Guillain-Barré syndrome, myasthenia gravis, and almost always in cervical spinal cord injury. Lawn and Wijdicks³⁶ advocated the use of a pulmonary function score, the sum of the vital capacity, maximum inspiratory pressure ($P_{I\max}$), and maximum expiratory pressure ($P_{E\max}$), to predict prolonged ventilation and the need for tracheostomy in patients with Guillain-Barré syndrome.

The patient developing slowly progressive respiratory impairment will experience dyspnea, increased use of accessory neck muscles, paradoxical breathing, cough after swallowing (indicative of aspiration), tachycardia, sweating, and inability to say more than a few words in a row. Some patients may complain of headaches or sleep disturbances or exhibit cognitive impairment associated with daytime hypercapnia with partial compensation. Those with generalized muscle weakness often experience orthopnea whereas patients with quadriplegia prefer the supine position because the diaphragm assumes a greater resting length and force generation.⁴

Serial measurements of forced vital capacity provide a simple and reliable means of assessing disease progression, and may predict difficulties with cough and secretion clearance³⁷ although as much as 50% reduction in respiratory muscle strength occurs before these values are reduced.³⁸ In general, patients with less than 10% of predicted (or <1 L) vital capacity have little tolerance off assisted ventilation.^{2,3} In patients with bulbar involvement who have difficulty holding a mouthpiece in place, a full face mask can be used to perform spirometry. Facemask measurements tend to produce vital capacity values larger than those recorded by mouthpiece, in some cases by as much as 1.2 L in the seated position.³⁹ A more useful guide to diaphragmatic weakness is the comparison of the vital capacity measured in the seated and supine positions. Reductions in vital capacity of 12% to 65% have been recorded in patients with neuromuscular disease^{3,4,40,41} (except in those with traumatic quadriplegia^{4,42,43}) on assuming the supine position.

Measurements of $P_{I\max}$ and $P_{E\max}$ are noninvasive and have established normal values for adults and children. They require, however, maximal effort and coordination, and are uncomfortable to perform. Low values may be difficult to interpret.^{44,45} Poor effort or air leaks around the mouthpiece give rise to submaximal measurements. Sniff esophageal pressure (Pes-sniff) and twitch transdiaphragmatic pressure (Pdi) tests are more precise but somewhat invasive (Fig. 32-3).⁴⁶⁻⁴⁸ Sniff nasal inspiratory force avoids the need for use of a mouthpiece and is easily performed by most patients, but may be tiring because of the need for repetition.^{49,50} A sniff nasal inspiratory force less than 40 cm H₂O has a sensitivity of 97% and specificity of 79% for predicting 6-month mortality in amyotrophic lateral sclerosis.⁵¹ The use of individual tests tends to overdiagnose respiratory muscle weakness by approximately 27%, while the combination of tests increases precision and reduces the diagnosis of respiratory muscle weakness by 19% to 56% (Fig. 32-4).⁵² Supine forced vital capacity, upright forced vital capacity, $P_{I\max}$, $P_{E\max}$, and Pdi-sniff are associated with tracheostomy-free survival, more so than Pa_{CO_2} (Fig. 32-5).⁴¹

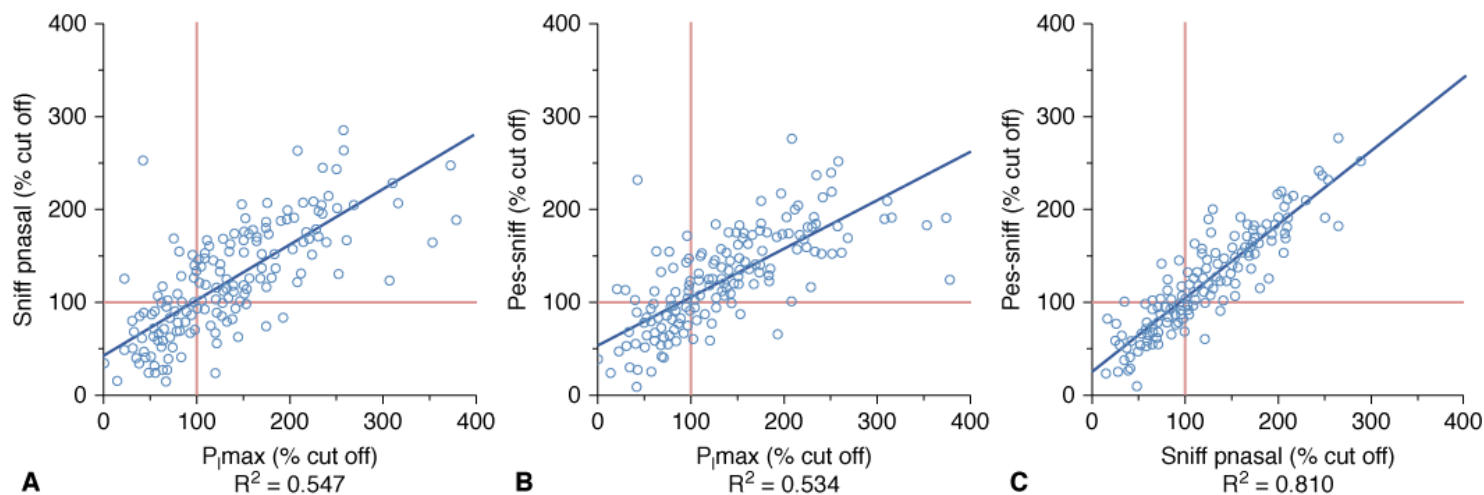
Figure 32-3



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Relationship between tests of respiratory muscle strength (RMS) and P_{CO_2} . Each graph shows earlobe blood gas P_{CO_2} in kPa on the y-axis and RMS on the x-axis. *Open symbols* represent the bulbar patients and *closed symbols* the limb patients. The *dotted lines* represent the lower and upper limits of normal for earlobe blood gas P_{CO_2} (4.8 to 6.0 kPa). *CMS Pdi*, transdiaphragmatic pressure after bilateral cervical magnetic phrenic nerve stimulation; *cough P_{gas}* , cough gastric pressure; *sniff Pdi*, maximal sniff transdiaphragmatic pressure; *Pes-sniff*, maximal sniff esophageal pressure; *SNP*, maximal sniff nasal pressure. (Used, with permission, from Lyall A, Donaldson N, Polkey MI, Leigh PN, Moxham J. Respiratory muscle strength and ventilatory failure in amyotrophic lateral sclerosis. *Brain*. 2001;124:2000–2013, with permission of Oxford University Press.)

Figure 32-4

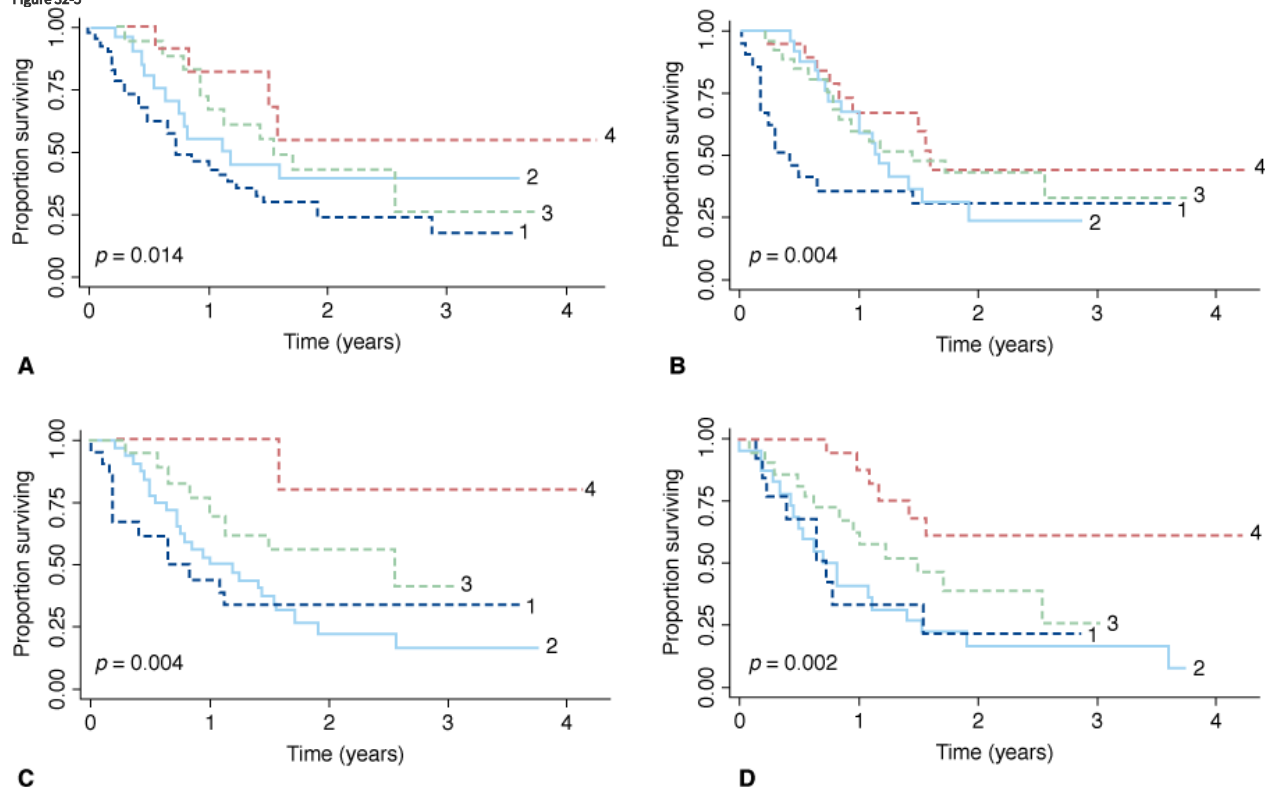


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Correlation between (A) maximum inspiratory pressure (P_{Imax}) and sniff nasal pressure (P_{nasal}), (B) P_{Imax} and sniff esophageal pressure (P_{es}), and (C) Sniff P_{nasal} and P_{es} -sniff. (Reproduced from Steier J, Kaul S, Seymour J, Jolley C, Rafferty G, Man W, Luo YM, Roughton M, Polkey MI, Moxham J. The value of multiple tests of respiratory muscle strength. *Thorax*. 2007;62:975–980, copyright 2007 with permission from BMJ Publishing Group Ltd.)

Figure 32-5



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Kaplan-Meier curves showing differences in survival based on values of supine forced vital capacity (FVC) (A), upright FVC (B), maximal inspiratory pressure (P_{Imax}) (C), and maximal expiratory pressure (P_{Emax}) (D). The categories of FVC are as follows: FVC $\leq 50\%$ (line 1); FVC = 51% to 65% (line 2); FVC = 66% to 80% (line 3); FVC $> 80\%$ (line 4). The P_{Imax} categories are as follows: P_{Imax} greater than -25 cm H₂O (line 1); P_{Imax} = -25 to -50 cm H₂O (line 2); P_{Imax} = -51 to -70 cm H₂O (line 3); P_{Imax} less than -70 cm H₂O (line 4). The P_{Emax} categories are as follows: P_{Emax} < 25 cm H₂O (line 1); P_{Emax} = 25 to 50 cm H₂O (line 2); P_{Emax} = 51 to 70 cm H₂O (line 3); P_{Emax} > 70 cm H₂O (line 4). The p values compare differences in survival between the four categories of each pulmonary test by the Wilcoxon test. (Used, with permission, from Schmidt et al.⁴¹)

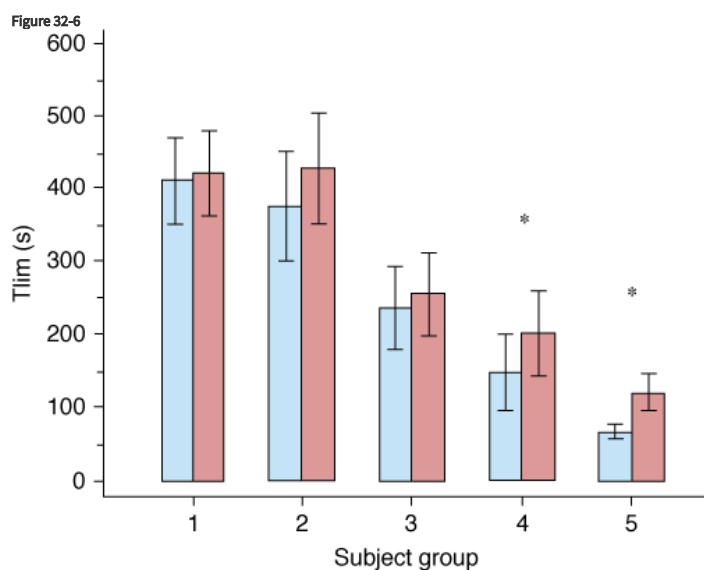
Table 32-2 lists physical and functional findings used in the evaluation of patients with impending respiratory failure secondary to neuromuscular disease.

Table 32-2: Indications for Initiating Mechanical Ventilation in Neuromuscular Diseases

Relative subjective symptoms (orthopnea, frequent arousals from sleep, morning headaches, daytime somnolence/napping, impaired cognitive function [memory, concentration], fatigue)
Cor pulmonale
Vital capacity <25% of predicted
Maximal inspiratory mouth pressure <25 cm H ₂ O
Nasal “sniff” pressure <25 cm H ₂ O
Changes in gas exchange indicating progressive respiratory failure:
Daytime PaCO ₂ >45 mm Hg
Sustained nocturnal hemoglobin desaturation (>5 minutes or >10% of total study time)
Recurrent acute respiratory failure episodes requiring intervention
Failure to respond to continuous positive pressure alone in patients with documented sleep apnea syndrome before neuromuscular weakness is detected

Ventilator Targets in Neuromuscular Disease

The chief goals are to reverse altered mental status, correct gas exchange, and improve quality of life (see [Table 32-2](#)). With the use of nighttime NIPPV, daytime hypercapnia in Duchenne muscular dystrophy is delayed by 4 to 5 years⁵³ and survival improves by 5 to 10 years.^{54,55} Application of a few hours of daytime NIPPV reduces breathlessness and increases endurance capacity in Duchenne muscular dystrophy more effectively than nocturnal NIPPV alone ([Fig. 32-6](#)).⁵⁵ Similarly, NIPPV improves survival and quality of life in amyotrophic lateral sclerosis, even in those with bulbar dysfunction.^{56,57}



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Endurance capacity (*Tlim*) in the five groups of patients with Duchenne muscular dystrophy. Columns represent mean values; errors bars represent 1 standard error of the mean (SEM). *Blue columns*: Evening *Tlim*; *red columns*: morning *Tlim*; **p* < 0.05 (paired t test, evening vs. morning). Group 1, no assisted ventilation and 24/24 hours normocapnia; group 2, nocturnal hypercapnia before the start of noninvasive positive-pressure ventilation (NIPPV) implementation; group 3, nocturnal NIPPV with no breathlessness; group 4, nocturnal NIPPV with breathlessness; group 5, full-time NIPPV. (Used, with permission, from Toussaint M, Soudon P, Kinnear W. Effect of non-invasive ventilation on respiratory muscle loading and endurance in patients with Duchenne muscular dystrophy. *Thorax*. 2008;63:430–434. Adapted by permission from BMJ Publishing Group Limited.)

Experience with Traditional Modes of Ventilation

Many patients with poliomyelitis were supported with negative-pressure devices such as oscillating beds ([Fig. 32-7](#)), iron lungs ([Fig. 32-8](#)), cuirasses, and “ponchos” ([Figs. 32-9A and B](#)). Subsequently, individuals with other neuromuscular disorders received tracheostomies for positive-pressure support ([Fig. 32-10](#)). Disadvantages of tracheostomies include the necessity of teaching the patient and caregivers how to maintain the tracheostomy, and complications such as stomal infections, airway erosions (from repeated suctioning), tracheal stenosis, and difficulty with speech and swallowing (that can be overcome with proper speech and bulbar training and use of a one-way speaking valve). Airway-vascular fistulas (with fatal hemorrhage) and tracheobronchomalacia are

infrequent but devastating complications that can occur even with uncuffed tracheostomies.⁵⁸ Tracheostomies can be performed with local anesthesia during NIPPV support in hypercarbic patients with neuromuscular disease (vital capacity <20% predicted).⁵⁹

Figure 32-7



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An oscillating bed used in patients with mild to moderate respiratory impairment.

Figure 32-8



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An iron lung (negative-pressure) ventilator.

Figure 32-9



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Two smaller negative-pressure ventilators. (A) Cuirass; (B) poncho type. (Used, with permission, from Respironics Inc., Murrysville, PA.)

Figure 32-10



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A poliomyelitis survivor receiving tracheostomy positive ventilation.

Noninvasive Positive-Pressure Ventilation

Not all patients elect to undergo tracheostomy for ventilation. The use of oral, lip (Fig. 32-11), nasal, or full-face (Figs. 32-12, 32-13, and 32-14) interfaces in neuromuscular disorders enables avoiding tracheostomies. They also have been used to wean patients off tracheostomy-assisted ventilation.³ An open mouthpiece circuit can be used in respiratory-dependent neuromuscular patients with a portable volume ventilator. Only volume-cycled ventilators with negative-pressure triggering should be used, however, because flow-triggered volume ventilators autocycle when used with an open-circuit mouth-positive system.^{3,60}

Figure 32-11



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An individual with late effects of poliomyelitis receiving mouth positive ventilation through a mouthpiece.

Figure 32-12



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A full-face mask used in noninvasive ventilation. (With permission from Hans Rudolph, Inc.)

Figure 32-13



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A silicone nasal interface used with noninvasive positive-pressure ventilation.

Figure 32-14

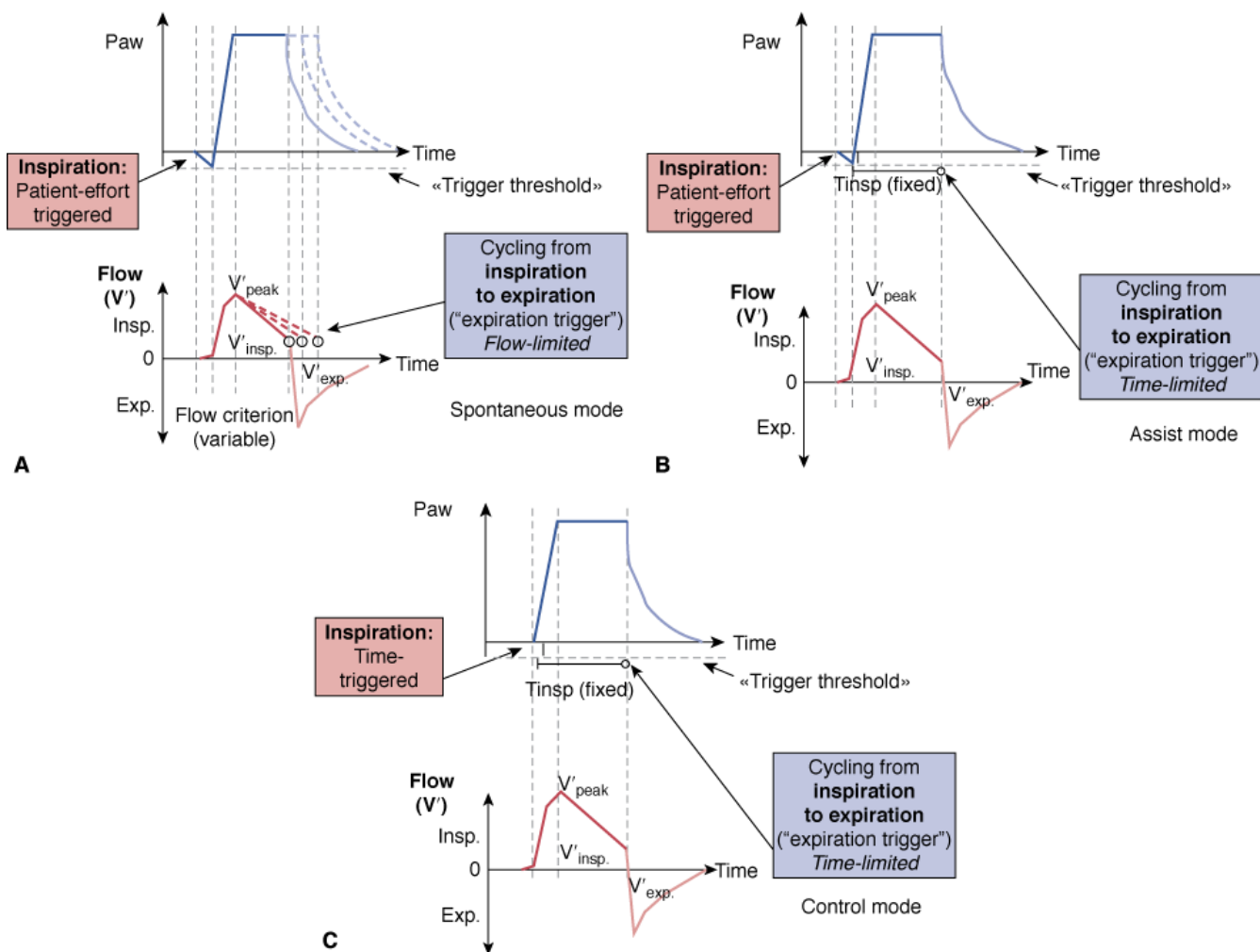


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Nasal adaptor ("pillows") used with noninvasive ventilation. (Used, with permission, from Swift™FX © ResMed 2012.)

Delivery of effective ventilation is dependent on many factors, including ventilator type, mode, settings, presence of leaks (Fig. 32-15), upper airway resistance, expiratory valve, type of triggering (Fig. 32-16), pressurization rate, cycling pattern, and positive end-expiratory pressure (PEEP) level.⁶⁰ Patients with increased respiratory elastance (e.g., scoliosis) are likely to benefit more from a volume-cycled ventilator than from a bilevel positive airway pressure machine. Volume-cycled ventilators have the advantage of delivering higher volumes and can provide air "stacking" to maximum insufflations via an oral interface.^{3,26} Leaks are best handled with pressure-targeted ventilators.⁶⁰

Figure 32-15

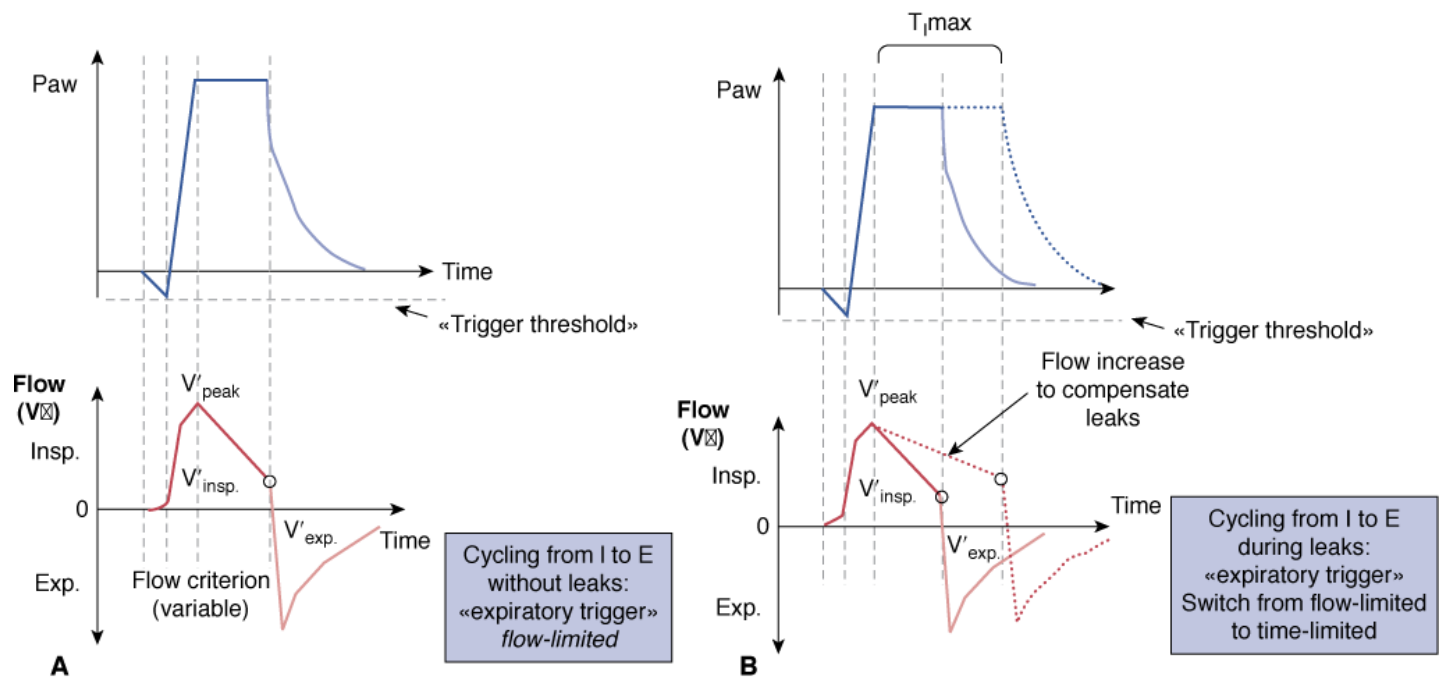


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Impact of leaks on inspiratory-to-expiratory (I to E) cycling during pressure support ventilation (S in bilevel devices) mode (A) without leaks and (B) with leaks. Note that during leaks the flow increases to compensate for them, prolonging the inspiratory time (*dashed lines*) and switching to time-limited if maximal inspiratory time ($T_{\text{I max}}$) is available. Interestingly, $T_{\text{I max}}$ is adjustable in most recent noninvasive ventilation devices. P_{aw} , pressure in the airways.

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Figure 32-16



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Different modes of inspiratory and expiratory triggering. **A**. Spontaneous mode. The patient controls the beginning and end of inspiration. Inspiration starts when the ventilator is triggered by the patient, and cycling from inspiration to expiration occurs when the inspiratory flow reaches a predetermined percentage of peak inspiratory flow. This mode is also called "pressure support" and, in some ventilators, "spontaneous (S) mode." **B**. Assisted mode. The patient controls the onset of inspiration but the end of inspiration is time triggered. When a backup respiratory rate (RR) is preset, this mode is called assist or control. In this last mode, if the patient's RR is lower than the preset ventilator backup RR, the system moves to control mode. **C**. Control mode. There is a preset automatic cycle based on time. The ventilator controls the beginning and end of inspiration and thus the RR. In some ventilators, this mode is also called "timed (T) mode." *Paw*, pressure in the airways. (Reproduced with permission from BMJ Publishing Group Ltd from Rabec C, Rodenstein D, Leger P, Rouault, Perrin C, Gonzalez-Bermejo, and on behalf of the SomnoNIV group. Ventilator modes and settings during non-invasive ventilation: effects on respiratory events and implications for their identification. *Thorax*. 2011;66:170–178.)

With a close working relationship between the patient and caregivers, machine tolerance can be achieved in virtually all patients with neuromuscular diseases, including those with hypercapnic amyotrophic lateral sclerosis.⁶¹ The prognosis in the latter group is particularly poor if they do not accept mask ventilation (death usually within less than a year). One study found a survival disadvantage in patients treated early with NIPPV for Duchenne muscular dystrophy,⁶² but this finding may have resulted from the exclusion of hypercapnic patients.

Current evidence concerning the benefits of NIPPV is consistent, suggesting relief of symptoms of hypoventilation in the short-term and, for the most part, improvement in survival^{56,57} and quality of life.^{47,56} Development of a home protocol in which oxygen desaturations are prevented or reversed using NIPPV and manual and mechanically assisted coughing techniques results in fewer and shorter hospitalizations than if such a protocol were not used.³ Early respiratory evaluation improves patient outcomes, particularly in patients without bulbar dysfunction, and increases survival time.⁵⁷ Bulbar involvement worsens survival and increases intolerance of NIPPV.^{56,57,61} Initiation of NIPPV in the hospital also improves tolerance through nocturnal adaption, and increases the likelihood of adhering to the home-ventilation program following discharge^{57,61}

Role for Newer Modes of Ventilation

Although pressure-cycled ventilation is the most common mode of providing ventilatory assistance in the long-term setting, patient-ventilation asynchrony sometimes occurs.⁶³ Proportional-assist ventilation improves patient-ventilatory synchrony by providing inspiratory flow and pressure in proportion to the patient's effort.⁶⁴ In a crossover study of fourteen patients with chronic ventilatory insufficiency (ten with restrictive thoracic disorders) comparing pressure-cycled with proportional-assist ventilation, both modes were shown to result in similar tolerance and were equally effective in reducing daytime hypercapnia and improving nocturnal saturation and symptoms.⁶⁵

Recently, average volume-assured pressure-support ventilation (AVAPS) (Fig. 32-17) has gained use. This mode guarantees any tidal volume compatible with the physician-set upper pressure limit. It applies the benefits of pressure control while avoiding its potential complications. It is an ideal mode for individuals with low respiratory compliance, such as those with morbid obesity, scoliosis, and atelectasis. AVAPS can also reduce the leak related to the interface and reduce the rate of asynchrony related to the underlying disease process. In the only randomized, crossover trial in patients with a chest wall disorder (obesity-

hypoventilation syndrome), the addition of AVAPS to bilevel pressure ventilation reduced hypercapnia as well as improved sleep quality and health-related quality of life.⁶⁶

Figure 32-17



B

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IPAP max

IPAP min

EPAP

Target V_T

Patient flow

A

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A. Average volume-assured pressure-support (AVAPS) ventilator with humidifier. **B.** Tracings taken from AVAPS mode. Based on a sensitivity algorithm performance, the device estimates the patient's tidal volume (V_T) with each breath and compares it with the target tidal flow/volume. If necessary, the algorithm slowly increases or decreases inspiratory pressure for each breath (0.5 to 1 cm H_2O /min) so as to achieve the preset V_T . (Image provided by Philips Respironics, Murrysville, PA.)

Role for Positive End-Expiratory Pressure

PEEP is used primarily to improve oxygenation in patients with severe and refractory hypoxemia. PEEP should recruit gas exchange units in neuromuscular weakness as most patients have reduced functional residual capacities and dependent alveolar units that fail to open during inspiration.

The most practical application of PEEP is in the form of the expiratory positive airway pressure delivered by bilevel positive airway pressure. Although pressure-cycled ventilators can compensate partially for mask leaks, excessive leaks can prolong the inspiratory phase and worsen intrinsic PEEP and patient-ventilator asynchrony. The latter problem can be corrected with the use of time-cycled expiratory triggers better than with flow-cycled expiratory triggers.⁶⁷ As in any patient with impaired left-ventricular function receiving high levels of PEEP, excess pressures can lead to a decrease in cardiac output, a potential problem in dystrophic cardiomyopathies.

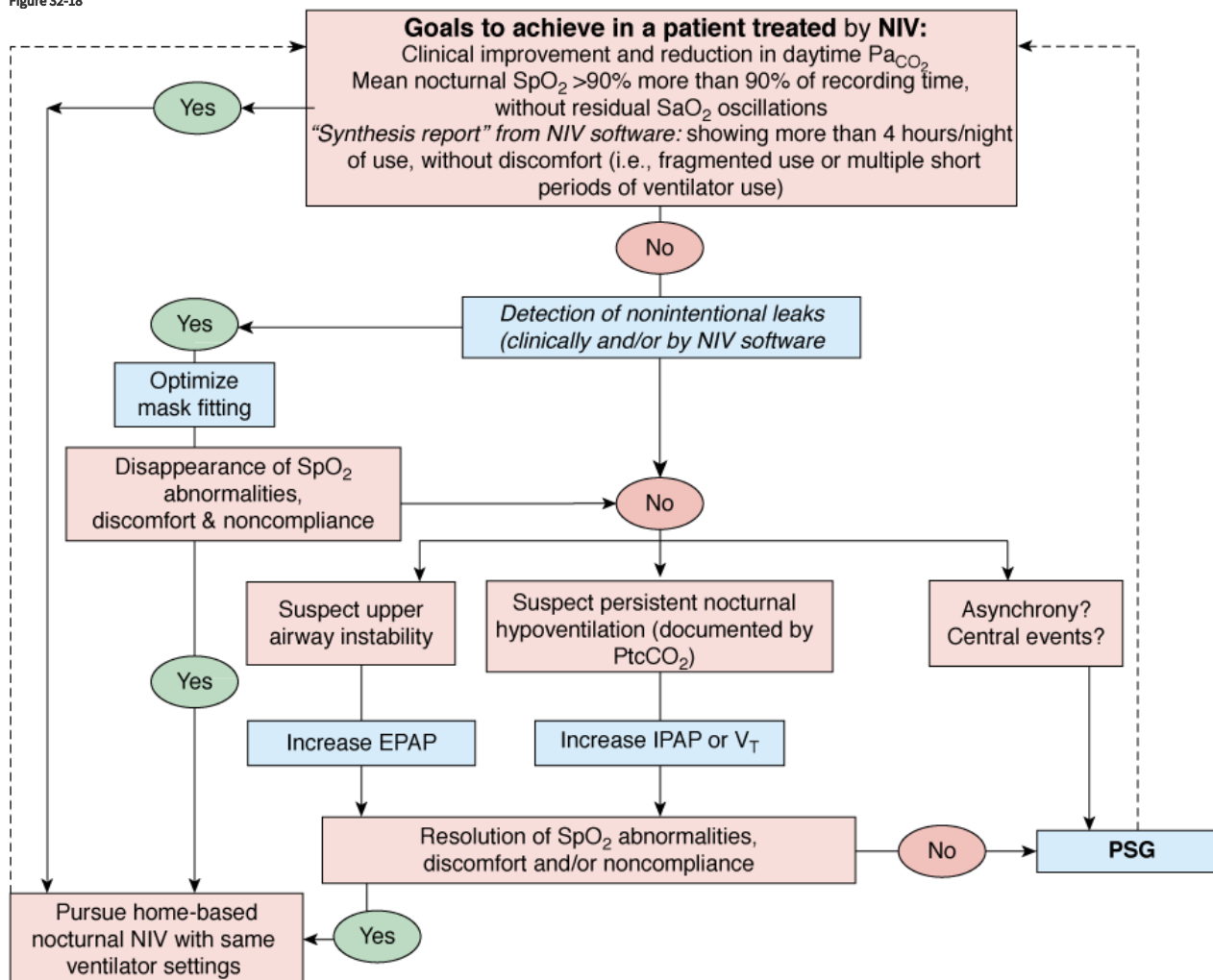
Selecting Ventilator Settings

For most patients who present with findings of respiratory impairment, as shown in Table 32-2, ventilator assistance usually is begun nocturnally. Bach³ recommends that most patients be started on a volume-cycled ventilator to achieve air “stacking” for maximum insufflations and for eventual daytime use. Occasional patients will benefit more from a pressure-cycled ventilator if they have glottic incompetence and cannot stack breaths.^{3,60} If they do use a pressure-cycled device, the pressure spans must be high enough to generate adequate volumes. The expiratory positive airway pressure should be set as low as possible (e.g., 2 cm H₂O) unless the patient has underlying lung disease or documented sleep apnea. The spontaneous-timed mode is recommended. Patients with oral leaks can be switched to a full-face mask or, if this is uncomfortable, to chin straps, or mouth or lip seals. Because such interfaces result in an open system, and patients can take in as much volume as they want, the set volume should be able to compensate for leaks (starting at 600 to 1000 mL, with eventual increases to 800 to 1500 mL).³

In patients with severe respiratory muscle weakness, patient-ventilator dyssynchrony may result in hypoventilation, particularly at night, when changes in the pattern of recruitment of the respiratory muscles may occur during different sleep stages and disrupt sleep. Settings of NIPPV chosen on an empirical basis while the patient is awake may not predict ventilator asynchrony during sleep, and improper titration of inspiratory support or PEEP may impede the trigger of the mechanical breath. These abnormalities can be reduced when a physiologic setting, based on recordings of inspiratory muscle efforts, is applied.⁶⁸

Noninvasive ventilation may be initiated in the hospital, a sleep laboratory, the outpatient office, or the patient's home, depending on the resources available. Ideally, the procedure should be initiated with the patient alert, under the guidance and supervision of experienced personnel. The goals should be to familiarize and encourage the patient with the technique and to establish initial ventilator settings (determined by optimizing both by physiology and patient comfort). Once gas-exchange and patient-comfort goals have been reached, transition to nocturnal use is initiated. Evaluations should be conducted using nocturnal oximetry and daytime arterial gases (Fig. 32-18). Polysomnography (and where available continuous percutaneous or end-tidal CO₂ monitoring) are recommended in patients in whom sleep hypoventilation is suspected.^{69,70,103}

Figure 32-18



Suggested algorithm for monitoring noninvasive ventilation (NIV) during sleep. *IPAP*, inspiratory positive airway pressure; Pa_{CO_2} , arterial carbon dioxide tension; *PtcCO₂*, transcutaneous pressure of carbon dioxide; *PSG*, polysomnography; *SaO₂*, arterial oxygen saturation; *SpO₂*, oxygen saturation measured by pulse oximetry; *V_T*, tidal volume. (Used, with permission, from Janssens J-P, Borel J-C, Pépin J-L, on behalf of the SomnoNIV group. Nocturnal monitoring of home non-invasive ventilation: the contribution of simple tools such as pulse oximetry, capnography, built-in ventilator software and autonomic markers of sleep fragmentation. *Thorax*. 2011;66(5):438–445. Adapted with permission from BMJ Publishing Group Limited.)

Vulnerability to Ventilator Complications

Many patients tend to have ventilation increased, especially during an acute respiratory infection, and neglect to restore their settings to their original values. Reducing the ventilation of chronically hyperventilated patients results in immediate dyspnea and demands to restore the previous ventilator settings.^{71,72} In contrast to tracheostomy-assisted ventilation, NIPPV provides an open system of ventilation with maintenance of more optimal levels of partial pressure of carbon dioxide (P_{CO_2}), allowing restoration of the ventilatory drive.³ The correction should be done with small decrements in tidal volume and/or respiratory rate over time. Periodic assessments of blood gases or end-tidal or rebreath P_{CO_2} values⁷³ should be done to optimize ventilation. Progressive weight loss in patients with chronic neuromuscular disorders may also necessitate a decrease in tidal volume requirements.

Adverse effects of the interface, although not life-threatening, can lead to serious discomfort and the possibility of discontinuation. They include skin and eye irritation, drying of mucus membranes, aerophagia, leaks around the nasal and full-face interfaces, and abdominal distension with colic.⁷⁴

Volume-cycled ventilation in control mode may result in a decrease in effective ventilation secondary to glottic closure.^{3,60,75,76} Mechanical dysfunction and power outages occur occasionally, leaving the ventilator-dependent patient vulnerable to acute or progressive respiratory failure. The physician may have to exert influence as the patient's advocate to persuade third-party payors to provide reimbursement for a backup ventilator.

Other respiratory aids such as rocking beds, pneumobelts, diaphragmatic pacing, and negative pressure ventilators are discussed elsewhere in this book. In general, they are not as effective as NIPPV, particularly in patients with severe scoliosis and bulbar dysfunction. Negative pressure ventilators and diaphragmatic pacers can promote obstructive apneas, whereas the disadvantages of diaphragmatic pacers also include their high cost, lack of alarms, and potential to fail.

Adjunctive Therapy

Glossopharyngeal Breathing

Originally taught to persons with poliomyelitis, glossopharyngeal breathing (GPB) may be useful in other neuromuscular diseases in maintaining ventilation. In individuals with weak inspiratory muscles and no tolerance off the ventilator, GPB is a nonmechanical form of breathing that can be handy in the case of sudden ventilator failure or to augment spontaneous breathing.⁷⁷ With each closure of the glottis (a “gulp”; 40 to 200 mL) can be inhaled and then exhaled to augment breathing. GPB can be used to stack breaths (up to 1.6 L) and promote cough.³

Assisted Coughing

Respiratory complications are the principal causes of morbidity and mortality in advanced neuromuscular diseases, in particular, inability to eliminate airway secretions because of ineffective coughing. Clearance of secretions with assisted coughing can prevent respiratory failure and delay or avoid the need for a tracheostomy. Effectiveness of secretion clearance depends on the peak expiratory flow rate. A peak cough flow of 160 to 270 L/min is an indication to provide manually or mechanically applied cough assistance.^{26,78} These values should be used as guidelines and not as absolute cutoffs because some patients may develop difficulty eliminating secretions despite higher peak flows.

Manually assisted coughing requires cooperation and coordination between the patient and caregiver. The patient achieves a maximal inspiratory capacity, preferably with use of a positive-pressure device (such as with a resuscitator bag, or ventilator, or mechanical insufflation–exsufflation device) and breath “stacking,” closes his or her glottis, and then “lets go” with the help of an externally applied upward and inward abdominal thrust.⁷⁹ A mouth pressure of 60 cm H₂O or more during relaxation at peak insufflation against a closed shutter usually indicates a recoil pressure adequate to generate a good cough flow.⁷⁹

Patients with quadriplegia can contract the clavicular portion of the pectoralis major to compress the rib cage during forced expiration and cough.⁸⁰ They have decreased ability to increase intrathoracic pressure, however, because of paralysis of expiratory rib cage and abdominal muscles. Electrical or magnetic stimulation can increase intrathoracic pressure inversely and may be used to augment dynamic airway compression and clearance of airway secretions.⁸¹

Mechanical Insufflation–Exsufflation Devices

A mechanical insufflation–exsufflation device is a positive-pressure generator that produces a deep inspiration followed by a powerful sucking action that achieves a high expiratory flow rate (see Fig. 32-19). Minimum pressures of 35 to 40 cm H₂O are required to expel secretions but should be increased gradually

over time according to the patient's comfort level.⁸² Such pressures would be required to overcome small lungs and stiff chest walls, as in scoliosis. The mechanical insufflation–exsufflation device can be used with tracheostomies and full-face masks, avoiding the mucosal trauma that results from suctioning. Frequency of use can vary depending on the amount of secretions produced. Many patients note that they can breathe more easily after using the mechanical insufflation–exsufflation device. These devices are useful in patients with most neuromuscular conditions, including amyotrophic lateral sclerosis.^{83,84} Children with neuromuscular disorders exhibit pressure-dependent short-term improvements in inspiratory and expiratory pressures and flows, associated with expired volume measured during the mechanical insufflation–exsufflation maneuver with good tolerance.⁸⁵ For optimal results, use of the mechanical insufflation–exsufflation device can be augmented with application of manual cough technique and proper positioning of the patient to promote drainage of airway secretions.

Figure 32-19

A



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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(A) A mechanical insufflation–exsufflation device shown ready to use with a full-face mask **(B)** shown in use through a tracheostomy. (Image provided by Philips Respironics, Murrysville, PA.)

The mechanical insufflation–exsufflation device can also function as an alternative to intermittent positive-pressure breathing. By increasing lung volume during assisted cough, it can generate a higher peak cough flow than unassisted cough. It can improve other respiratory variables, including oxygen saturation and dyspnea, in neuromuscular diseases.⁸⁶

Complications with mechanical insufflation–exsufflation devices are rare and include nausea, bradycardia, tachycardia, and abdominal distension. The most common reason for intolerance of the mechanical insufflation–exsufflation device is inadequate caregiver training regarding its application. As with any respiratory device used for life support, education of the patient and caregivers is important to ensure best results.

Bronchoscopy is useful only in selected cases of atelectasis caused by mucus plugging and should be tried after other methods of secretion removal have failed.

Bulbar Dysfunction and Nutritional Support

Patients with severe dysphagia require a gastrostomy for continued nutritional support. The gastrostomy is often placed in patients with an artificial airway under general anesthesia. Avoiding intubation and general anesthesia decreases the risk of respiratory complications and can prolong noninvasive ventilator management. Bach et al⁸⁷ reported placement of gastrostomies in sixty-two patients with neuromuscular disease while receiving NIPPV supplemented by mechanically assisted coughing. Normal gas exchange was able to be maintained and there were no complications.

As with any patient in a critical care setting, the patient with neuromuscular disease must be guarded against overfeeding. Because of progressive muscle wasting, caloric replacement based on ideal body weight (assuming intact muscle mass) can lead to rapid weight gain, reductions in lung volume and respiratory compliance, and an increase in work of breathing.

Timing of Transfer from Intensive Care Unit to the Community

Patients may be hospitalized for a number of acute consequences of respiratory muscle weakness, including pneumonia, atelectasis, hypoventilation, and sleep apnea. After the complication has resolved, the patient must be prepared for discharge into the community using a multidisciplinary approach incorporating potential caregivers in the planning. During this period, there should be clarification of the patient's advance directives, particularly with regard to interventions for long-term survival. If the patient is supported with NIPPV, attention must be given to maintenance of appropriate ventilator settings, the oral–nasal interface, seating, mobility, and nutrition (including gastrostomy or jejunostomy feeding).^{3,60} Caregivers must be trained in techniques of manual and mechanical cough assistance. Only if such measures are unsuccessful in clearing secretions (as in severe bulbar involvement with amyotrophic lateral sclerosis or postpolio syndrome) should tracheostomies be resorted to. As yet, professional organizations do not provide guidelines for extubating patients with neuromuscular diseases and critical care neuropathy-myopathy. The use of assisted coughing can be used to transition from tracheostomy-assisted ventilation to noninvasive ventilation in suitable patients who meet certain criteria, the main ones including being oxygenation and absence of hypercapnia, being fully alert and cooperative, and a peak expiratory flow of at least 160 L/m.³ Many patients have expressed a preference to NIPPV, even after many years of having lived with tracheostomies.

Long-Term Survival and Quality of Life

Home NIPPV improves survival in amyotrophic lateral sclerosis^{61,88,89} and Duchenne muscular dystrophy,^{90–92} as well as quality of life in most patients.^{47,56,61} Yet clinical practice remains variable among physicians,^{93–96} most likely because of a perception that mechanical ventilation may prolong suffering in progressive disease. In general, health care workers tend to underestimate quality-of-life scores of life satisfaction.⁹⁷

Patients appreciate having meaningful discussions about assisted ventilation throughout the course of their illness, and they should begin before the onset of respiratory impairment. Care should be conducted in a multidisciplinary setting involving neurologists, pulmonologists, psychologists, social workers, physical and occupational therapists, speech pathologists, nutritionists, and respiratory therapists.^{98–102} In some respects, working closely with the family and other caregivers is even more vital than working with the patient, but the workload associated with supporting home-ventilated patients can be considerable. Despite patients and caregivers receiving competency training before discharge, regular reeducation is required. Urgent calls to a dedicated respiratory support system may be reduced by a more flexible problem-solving approach. Reports in which no technical fault is found may indicate declining health and require clinical reevaluation.¹⁰²

Important Unknowns and the Future

Although it is agreed that NIPPV is the preferred means of managing ventilatory impairment and failure in neuromuscular disease, many patients have difficulty in tolerating mask ventilation. Inability or unwillingness to use the ventilator detracts from quality of life as well as survival. Glottic closure, patient–ventilator asynchrony leading to increased work of breathing in those patients with remaining respiratory muscle function, a sensation of suffocation, and facial pressure points and abrasions contribute to this lack of success. Increased use of spontaneous-timed ventilator mode ventilation and of oral or lip seals may circumvent some of these problems. Well-designed, randomized, controlled trials and cost-effectiveness studies would help to increase the acceptability of NIPPV at an earlier stage of respiratory impairment, enabling the patient to increase its use with disease progression. Ethical issues also present problems, for example, by not comparing hospitalization rates of an untreated group with the hospitalizations of patients who have avoided tracheostomy through the use of NIPPV, respiratory muscle aids, and cough-assist techniques. Such studies also would increase physician acceptance of the use of NIPPV as a routine means of improving quality of life and survival. A change in physician attitudes will lead to greater acceptance among patients who, unfortunately, still face indifferent and even negative attitudes concerning life support.

Summary and Conclusion

Noninvasive ventilation is now the accepted form of respiratory support for people with neuromuscular and chest wall disorders. Improvements in ventilator modes, interfaces, and adjunctive secretion-removal techniques have facilitated care and increased the quality of life in such individuals. An interdisciplinary approach aimed at not only the patient but also the family and other caregivers is essential for coping with the rigors of providing care under potentially

stressful conditions. Close involvement by third-party payors and private community-based and governmental agencies is also crucial to ease the cost of home care.

References

1. Affeldt JE, Bower AG, Dail CW. Prognosis for respiratory recovery in severe poliomyelitis. *Arch Phys Med Rehabil.* 1957;38:290–295. [[PubMed: 13412279](#)]

2. Baydur A, Layne E, Aral H, et al. Long term non-invasive ventilation in the community for patients with musculoskeletal disorders: 46 year experience and review. *Thorax.* 2000;55:4–11. [[PubMed: 10607795](#)]

3. Bach J. *Management of Patients with Neuromuscular Disease.* Philadelphia, PA: Hanley and Belfus; 2004:211–308.

4. Baydur A, Adkins RH, Milic-Emili J. Lung mechanics in individuals with spinal cord injury: effects of injury level and posture. *J Appl Physiol.* 2001;90:405–411. [[PubMed: 11160035](#)]

5. Gibson GJ, Pride NB, Newsom Davis J, et al. Pulmonary mechanics in patients with respiratory muscle weakness. *Am Rev Respir Dis.* 1977;115:389–395. [[PubMed: 842952](#)]

6. Loh L, Hughes JMB, Newsom Davis J. Gas exchange problems in bilateral diaphragm paralysis. *Bull Eur Physiopathol Respir.* 1979;15:137–141. [[PubMed: 122063](#)]

7. Bye PTP, Ellis ER, Issa FG, et al. Respiratory failure and sleep in neuromuscular disease. *Thorax.* 1990;45:241–247. [[PubMed: 2113317](#)]

8. Smith PEM, Edwards RHT, Calverley PMA. Ventilation and breathing pattern during sleep in Duchenne muscular dystrophy. *Chest.* 1989;96:1346–1351. [[PubMed: 2582842](#)]

9. Farkas GA, McCormick KM, Gosselin LE. Episodic hypoxia exacerbates respiratory muscle dysfunction in DMD^{MDX} mice. *Muscle Nerve.* 2007;36:708–710. [[PubMed: 17654561](#)]

10. Laghi F, Tobin MJ. Disorders of the respiratory muscles. *Am J Respir Crit Care Med.* 2003;168:10–48. [[PubMed: 12826594](#)]

11. Stübgen J-P, Schultz C. Lung and respiratory muscle function in facioscapulohumeral muscular dystrophy. *Muscle Nerve.* 2009;39:729–734. [[PubMed: 23268520](#)]

12. Toussaint M, Steens M, Soudon P. Lung function accurately predicts hypercapnia in patients with Duchenne muscular dystrophy. *Chest.* 2007;131:368–375. [[PubMed: 17296635](#)]

13. Redding GJ, Okamoto GA, Guthrie RD, et al. Sleep patterns in nonambulatory boys with Duchenne muscular dystrophy. *Arch Phys Med Rehabil.* 1985;66:818–821. [[PubMed: 4074115](#)]

14. Bourke SC, Gibson GJ. Sleep and breathing in neuromuscular disease. *Eur Respir J.* 2002;19:1194–1201. [[PubMed: 12108875](#)]

15. Khan Y, Heckmatt JZ. A double blind cross over trial of theophylline prophylaxis for sleep hypoxaemia in Duchenne muscular dystrophy. *Neuromuscul Disord.* 1997;7:75–80. [[PubMed: 9131646](#)]

16. Piper AJ, Sullivan CE. Effects of long-term nocturnal nasal ventilation on spontaneous breathing during sleep in neuromuscular and chest wall disorders. *Eur Respir J.* 1996;19:1515–1522.

17. Diaz CE, Deoras KS, Allen JL. Chest wall motion before and after mechanical ventilation in children with neuromuscular disease. *Pediatr Pulmonol.* 1993;16:89–95. [[PubMed: 8367222](#)]

18. Estenne M, Heilporn A, Delhez L, et al. Chest wall stiffness in patients with chronic respiratory muscle weakness. *Am Rev Respir Dis.* 1983;128:1002–1007. [[PubMed: 6228174](#)]

19. De Troyer A, Borenstein S, Cordier R. Analysis of lung volume restriction in patients with respiratory muscle weakness. *Thorax.* 1980;35:603–610.

20. Baydur A, Swank, Stiles CM, Sassoon CSH. Respiratory elastic load compensation in anesthetized patients with kyphoscoliosis. *J Appl Physiol.* 1989;67:1024–1031. [[PubMed: 2793696](#)]

21. Baydur A, Carlson M. Immediate response to inspiratory resistive loading in anesthetized patients with kyphoscoliosis: spirometric and neural effects. *Lung*. 1996;174:99–118. [[PubMed: 8919433](#)]

22. Guérin C, Vincent B, Petitjean T, et al. The short-term effects of intermittent positive pressure breathing treatments on ventilation in patients with neuromuscular disease. *Respir Care*. 2010;55:866–872. [[PubMed: 23268345](#)]

23. Sinha R, Bergofsky EH. Prolonged alteration of lung mechanics in kyphoscoliosis by positive pressure hyperinflation. *Am Rev Respir Dis*. 1972;106:47–57. [[PubMed: 4556537](#)]

24. Vincken W, Elleker MG, Cosio MG. Determinants of respiratory muscle weakness in stable chronic neuromuscular disorders. *Am J Med*. 1987;82:53–58. [[PubMed: 3799694](#)]

25. Bach JR, Saporito LR. Criteria for extubation and tracheostomy tube removal for patients with ventilatory failure: a different approach to weaning. *Chest*. 1996;110:1566–1571. [[PubMed: 8989078](#)]

26. Tzeng AC, Bach JR. Prevention of pulmonary morbidity for patients with neuromuscular disease. *Chest*. 2000;118:1390–1396. [[PubMed: 11083691](#)]

27. Begin P, Mathieu J, Almirall J, et al. Relationship between chronic hypercapnia and inspiratory muscle weakness in myotonic dystrophy. *Am J Respir Crit Care Med*. 1997;156:133–139. [[PubMed: 9230737](#)]

28. Veale D, Cooper BG, Gilmartin JJ, et al. Breathing pattern awake and asleep in patients with myotonic dystrophy. *Eur Respir J*. 1995;8:815–818. [[PubMed: 7656955](#)]

29. Piper A. Sleep abnormalities associated with neuromuscular disease: pathophysiology and evaluation. *Semin Respir Crit Care Med*. 2002;23:211–219. [[PubMed: 16088613](#)]

30. Fergusson KA, Strong MJ, Ahmad D, et al. Sleep-disordered breathing in amyotrophic lateral sclerosis. *Chest*. 1996;110:664–669.

31. Finnimore AJ, Jackson RV, Morton A, Lynch E. Sleep hypoxia in myotonic dystrophy and its correlation with awake respiratory function. *Thorax*. 1994;49:66–70. [[PubMed: 8153943](#)]

32. White JES, Drinnan MJ, Smithson AJ, et al. Respiratory muscle activity and oxygenation during sleep in patients with muscle weakness. *Eur Respir J*. 1995;8:807–814. [[PubMed: 7656954](#)]

33. Gay PC, Westbrook PR, Daube JR, et al. Effects of alterations in pulmonary function and sleep variables on survival in patients with amyotrophic lateral sclerosis. *Mayo Clin Proc*. 1991;66:686–694. [[PubMed: 2072756](#)]

34. Khan Y, Heckmatt JZ. Obstructive apneas in Duchenne muscular dystrophy. *Thorax*. 1994;49:157–161. [[PubMed: 8128406](#)]

35. Baydur A. Respiratory muscle strength and control of ventilation in patients with neuromuscular disease. *Chest*. 1991;99:330–338. [[PubMed: 1989791](#)]

36. Lawn ND, Wijedicks EF. Post-intubation pulmonary function test in Guillain-Barré syndrome. *Muscle Nerve*. 2000;23:613–616. [[PubMed: 10716773](#)]

37. Phillips MF, Quinlivan RCM, Edwards RHT, Calverley PMA. Changes in spirometry over time as a prognostic marker in patients with Duchenne muscular dystrophy. *Am J Respir Crit Care Med*. 2001;164:2191–2194. [[PubMed: 11751186](#)]

38. Braun NMT, Arora NS, Rochester DF. Respiratory muscle and pulmonary function in polymyositis and other proximal myopathies. *Thorax*. 1983;38:616–623. [[PubMed: 6412385](#)]

39. Khaliq AA, Smith IE. Facemask spirometry in patients with bulbar motor neuron disease (S83). *Thorax*. 2009;64(Suppl IV):A44.

40. Newsom-Davis J, Goldman M, Loh L, et al. Diaphragm function and alveolar hypoventilation. *Q J Med*. 1976;45:87–100.

41. Schmidt EP, Drachman DB, Wiener CM, et al. Pulmonary predictors of survival in amyotrophic lateral sclerosis: use in clinical trial design. *Muscle Nerve*. 2006;33:127–132. [[PubMed: 16258948](#)]

42. Estenne M, De Troyer A. Mechanism of the postural dependence of vital capacity in tetraplegic subjects. *Am Rev Respir Dis*. 1987;135:367–371. [[PubMed: 3813198](#)]

43. Maeda CJ, Baydur A, Waters RL, et al. The effect of the halovest and body position on pulmonary function in quadriplegia. *J Spinal Disord*. 1990;3:47–51. [[PubMed: 2134410](#)]

-
44. ATS/ERS. Joint statement on respiratory muscle testing. *Am J Respir Crit Care Med*. 2002;166:518–524.
-
45. Heritier F, Rahm F, Pasche P, et al. Sniff nasal inspiratory pressure. A noninvasive assessment of inspiratory muscle strength. *Am J Respir Crit Care Med*. 1994;150:1678–1683. [[PubMed: 7952632](#)]
-
46. Stefanutti D, Benoist MR, Scheinmann P, et al. Usefulness of sniff nasal pressure in patients with neuromuscular or skeletal disorders. *Am J Respir Crit Care Med*. 2000;162:1507–1511. [[PubMed: 11029369](#)]
-
47. Lyall RA, Donaldson N, Polkey MI, et al. Respiratory muscle strength and ventilatory failure in amyotrophic lateral sclerosis. *Brain*. 2001;124:2000–2013. [[PubMed: 11571218](#)]
-
48. DePalo VA, McCool FD. Respiratory muscle evaluation of the patient with neuromuscular disease. *Semin Respir Crit Care Med*. 2002;23:201–209. [[PubMed: 16088612](#)]
-
49. Lofaso F, Nicot F, Lejaille M, et al. Sniff nasal inspiratory pressure: what is the optimal number of sniffs? *Eur Respir J*. 2006;27:980–982. [[PubMed: 16455823](#)]
-
50. Fitting J-W. Sniff nasal inspiratory pressure: simple or too simple? *Eur Respir J*. 2006;27:881–886. [[PubMed: 16707389](#)]
-
51. Morgan RK, McNally S, Alexander M, et al. Use of sniff nasal-inspiratory force to predict survival in amyotrophic lateral sclerosis. *Am J Respir Crit Care Med*. 2005;171:268–274.
-
52. Steier J, Kaul S, Seymour J, et al. The value of multiple tests of respiratory muscle strength. *Thorax*. 2007;62:975–980. [[PubMed: 17557772](#)]
-
53. Toussaint M, Steens M, Wasteels G, et al. Diurnal ventilation via mouthpiece: survival in end-stage Duchenne patients. *Eur Respir J*. 2006;28:549–555. [[PubMed: 16870671](#)]
-
54. Toussaint M, Chatwin M, Soudon P. Mechanical ventilation in Duchenne patients with chronic respiratory insufficiency: clinical implications of 20 years published experience. *Chron Respir Dis*. 2007;4:167–177. [[PubMed: 17711917](#)]
-
55. Toussaint M, Soudon P, Kinnear W. Effect of non-invasive ventilation on respiratory muscle loading and endurance in patients with Duchenne muscular dystrophy. *Thorax*. 2008;63:430–434. [[PubMed: 18057095](#)]
-
56. Bourke SC, Tomlinson M, Williams TL, et al. Effects of non-invasive ventilation on survival and quality of life in patients with amyotrophic lateral sclerosis: a randomised controlled trial. *Lancet Neurol*. 2006;5:140–147.
-
57. Farrero E, Prats E, Povedano M, et al. Survival in amyotrophic lateral sclerosis with home mechanical ventilation: the impact of systematic respiratory assessment and bulbar involvement. *Chest*. 2005;127:2132–2138. [[PubMed: 15947331](#)]
-
58. Baydur A, Kanel G. Tracheobronchomalacia and tracheal hemorrhage in patients with Duchenne muscular dystrophy receiving long-term ventilation with uncuffed tracheostomies. *Chest*. 2003;123:1307–1311. [[PubMed: 12684330](#)]
-
59. Orlikowski D, Prigent H, Gonzalez-Bermejo J, et al. Noninvasive ventilation as an alternative to endotracheal intubation during tracheotomy in advanced neuromuscular disease. *Respir Care*. 2007;52:1728–1733. [[PubMed: 18028563](#)]
-
60. Rabec C, Rodenstein D, Leger P, et al. Ventilator modes and settings during non-invasive ventilation: effects on respiratory events and implications for their identification. *Thorax*. 2011;66:170–178. [[PubMed: 20947891](#)]
-
61. Aboussouan LS, Khan SU, Meeker DP, et al. Effect of noninvasive positive-pressure ventilation on survival in amyotrophic lateral sclerosis. *Ann Intern Med*. 1997;127:450–453. [[PubMed: 9313002](#)]
-
62. Raphael JC, Chevret S, Chastang C, et al. Randomized trial of preventive nasal ventilation in Duchenne muscular dystrophy. French Multicentre Cooperative Group on Home Mechanical Ventilation Assistance in Duchenne de Boulogne Muscular Dystrophy. *Lancet*. 1994;343:1600–1604. [[PubMed: 7911921](#)]
-
63. Jubran A, Van de Graaff WB, Tobin MJ. Variability of patient-ventilator interaction with pressure support ventilation in patients with chronic obstructive pulmonary disease. *Am J Respir Crit Care Med*. 1995;152:129–136. [[PubMed: 7599811](#)]
-
64. Younes M. Proportional assist ventilation, a new approach to ventilatory support: theory. *Am Rev Respir Dis*. 1992;145:114–120. [[PubMed: 1731573](#)]
-

65. Winck JC, Vitacca M, Morais A, et al. Tolerance and physiologic effects of nocturnal mask pressure support vs proportional assist ventilation in chronic ventilatory failure. *Chest*. 2004;126:382–388. [PubMed: 15302721]
66. Storre JH, Seuthe B, Fiechter R, et al. Average volume-assured pressure support in obesity hypoventilation: a randomized crossover trial. *Chest*. 2006;130:815–821. [PubMed: 16963680]
67. Calderini E, Confalonieri M, Puccio PG, et al. Patient-ventilator asynchrony during noninvasive ventilation: the role of the expiratory trigger. *Intensive Care Med*. 1999;25:662–667. [PubMed: 10470568]
68. Fanfulla F, Delmastro M, Berardinelli A, et al. Effects of different ventilator settings on sleep and inspiratory effort in patients with neuromuscular disease. *Am J Respir Crit Care Med*. 2005;172:619–624. [PubMed: 15961699]
69. American Thoracic Society Statement. Respiratory care of the patient with Duchenne muscular dystrophy. *Am J Respir Crit Care Med*. 2004;170:456–465.
70. Perrin C, Unterborn JN, Ambrosio CD, et al. Pulmonary complications of chronic neuromuscular diseases and their management. *Muscle Nerve*. 2004;29:5–27. [PubMed: 14694494]
71. Bach JR, Haber II, Wang TG, et al. Alveolar ventilation as a function of ventilatory support method. *Eur J Phys Rehabil Med*. 1995;5:80–84.
72. Manning HL, Shea SA, Schwartzstein RM, et al. Reduced tidal volume increases air hunger at fixed pCO₂ in ventilated quadriplegics. *Respir Physiol*. 1992;90:9–30.
73. Hackney JD, Seares CH, Collier CR. Estimation of arterial CO₂ tension by rebreathing technique. *J Appl Physiol*. 1958;12:425–430. [PubMed: 13525305]
74. Leger P, Langevin B, Guez A, et al. What to do when nasal ventilation fails for neuromuscular patients? *Eur Respir Rev*. 1993;3:279–283.
75. Jounieaux V, Aubert G, Dury M, et al. Effects of nasal positive-pressure hyperventilation on the glottis in normal awake subjects. *J Appl Physiol*. 1995;79:176–185. [PubMed: 7559217]
76. Parreira VF, Jounieaux V, Aubert G, et al. Nasal two-level positive-pressure ventilation in normal subjects. *Am J Respir Crit Care Med*. 1996;153:1616–1623. [PubMed: 8630611]
77. Dail C, Rodgers M, Guess V, et al. *Glossopharyngeal Breathing*. Downey, CA: Rancho Los Amigos Department of Physical Therapy; 1979.
78. Bach JR, Smith WH, Michaels J, et al. Airway secretion clearance by mechanical exsufflation for post-poliomyelitis ventilator-assisted individuals. *Arch Phys Med Rehabil*. 1993;74:170–177. [PubMed: 8431102]
79. Kang SW, Bach JR. Maximum insufflation capacity. *Chest*. 2000;118:213–219.
80. De Troyer A, Estenne M, Heilporn A. Mechanism of active expiration in tetraplegic subjects. *N Engl J Med*. 1986;314:740–744.
81. Estenne M, Pinet C, De Troyer A. Abdominal muscle strength in patients with tetraplegia. *Am J Respir Crit Care Med*. 2000;161:707–712. [PubMed: 10712311]
82. Miske L, Hickey EM, Kolb SM, et al. Use of the mechanical in-exsufflator in pediatric patients with neuromuscular disease and impaired cough. *Chest*. 2004;125:1406–1412. [PubMed: 15078753]
83. Bach JR, Ishikima Y, Kim H. Prevention of pulmonary morbidity for patients with Duchenne muscular dystrophy. *Chest*. 1997;112:1024–1028. [PubMed: 9377912]
84. Sancho J, Servera E, Diaz J, et al. Efficacy of mechanical insufflation-exsufflation in medically stable patients with amyotrophic lateral sclerosis. *Chest*. 2004;125:1400–1405. [PubMed: 15078752]
85. Fauroux B, Guillemot N, Aubertin G, et al. Physiologic benefits of mechanical insufflation-exsufflation in children with neuromuscular disorders. *Chest*. 2008;133:161–168. [PubMed: 18071020]
86. Winck JC, Goncalves MR, Lourenco C, et al. Effects of mechanical insufflation-exsufflation on respiratory parameters for patients with chronic airway secretions encumbrance. *Chest*. 2004;126:774–780. [PubMed: 15364756]

87. Bach JR, Gonzalez M, Sharma A, et al. Open gastrostomy for noninvasive ventilation users with neuromuscular disease. *Am J Phys Med Rehabil.* 2010;89:1–6. [[PubMed: 20026942](#)]

88. Kleopa KA, Sherman M, Neal B, et al. Bipap improves survival and rate of pulmonary function decline in patients with ALS. *J Neurol Sci.* 1999;164:82–88. [[PubMed: 10385053](#)]

89. Pinto AC, Evangelista T, Carvalho MA, et al. Respiratory assistance with a noninvasive ventilator (BiPAP) in MND/ALS patients: survival rates in a controlled trial. *J Neurol Sci.* 1995;129:19–26. [[PubMed: 7595610](#)]

90. Simonds AK, Muntoni F, Heather S, et al. Impact of nasal ventilation on survival in hypercapnic Duchenne muscular dystrophy. *Thorax.* 1998;53:949–952. [[PubMed: 10193393](#)]

91. Jeppesen J, Green A, Steffensen BF, Rahbek J. The Duchenne muscular dystrophy population in Denmark, 1997–2001: prevalence, incidence and survival in relation to the introduction of ventilator use. *Neuromuscul Disord.* 2003;13:804–812. [[PubMed: 14678803](#)]

92. Eagle M, Baudoin SV, Chandler C, et al. Survival in Duchenne muscular dystrophy: improvements in life expectancy since 1967 and the impact of home nocturnal ventilation. *Neuromuscul Disord.* 2003;13:804–812.

93. Bourke SC, Shaw PJ, Williams TL, et al. Non-invasive ventilation in motor neurone disease: current UK practice. *Amyotroph Lateral Scler Other Motor Neuron Disord.* 2002;3:145–149. [[PubMed: 12495576](#)]

94. Cedarbaum JM, Stambler N. Disease status and use of ventilatory support by ALS patients. *Amyotroph Lateral Scler Other Motor Neuron Disord.* 2001;2:19–22. [[PubMed: 11465928](#)]

95. Hardart MKM, Burns J, Truog RD. Respiratory support in spinal muscular atrophy type I: a survey of physician practices and attitudes. *Pediatrics.* 2002;110:24–30.

96. Kampelmacher MJ, Gaytant MA, Westerman EJA. Home mechanical ventilation in patients with neuromuscular diseases. ISNO (Dutch Neuromuscular Research Centre). <http://www.isno.nl>. Accessed September 1, 2010.

97. Bach JR, Campagnolo DI, Hoeman S. Life satisfaction of individuals with Duchenne muscular dystrophy using long-term mechanical ventilatory support. *Am J Phys Med Rehabil.* 1991;70:129–135. [[PubMed: 2039614](#)]

98. Traynor BJ, Alexander M, Corr B, et al. Effect of a multidisciplinary amyotrophic lateral sclerosis (ALS) clinic on ALS survival: a population study, 1996–2000. *J Neurol Neurosurg Psychiatry.* 2003;74:1258–1261. [[PubMed: 12933930](#)]

99. Gilgoff I, Prentice W, Baydur A. Patient and family participation in the management of respiratory failure in Duchenne's muscular dystrophy. *Chest.* 1989;95:519–524. [[PubMed: 2920577](#)]

100. Baydur A, Gilgoff I, Prentice W, et al. Decline in respiratory function and experience with long-term assisted ventilation in advanced Duchenne's muscular dystrophy. *Chest.* 1990;97:884–889. [[PubMed: 2182299](#)]

101. Bradley MD, Orrell RW, Clarke J, et al. Outcome of ventilatory support for acute respiratory failure in motor neurone disease. *J Neurol Neurosurg Psychiatry.* 2002;72:752–756. [[PubMed: 12023419](#)]

102. Chatwin M, Heather S, Hanak A, et al. Analysis of home support and ventilator malfunction in 1,211 ventilator-dependent patients. *Eur Respir J.* 2010;35:310–316. [[PubMed: 19643945](#)]

103. Janssens J-P, Borel J-C, Pépin J-L, on behalf of the SomnoNIV group. Nocturnal monitoring of home non-invasive ventilation: the contribution of simple tools such as pulse oximetry, capnography, built-in ventilator software and autonomic markers of sleep fragmentation. *Thorax.* 2011;66(5):438–445. [[PubMed: 20971980](#)]

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